

Hierarchical Clustering — Agglomerative Practical

Student practical guide: bottom-up hierarchical clustering using a small customer dataset.

1. Definition

Hierarchical Clustering is an unsupervised learning technique that creates a hierarchy of clusters. In **Agglomerative Hierarchical Clustering**, every observation starts as its own cluster and the closest clusters are repeatedly merged until one cluster remains.

2. Dataset

Create a file named **customers.csv**:

```
Customer,Annual_Income,Spending_Score
A,20,80
B,22,85
C,25,78
D,60,30
E,62,25
F,65,28
```

3. Libraries

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage, fcluster
```

4. Algorithm workflow

1. Calculate distances between observations.
2. Treat each observation as a separate cluster.
3. Find the closest clusters.
4. Merge them.
5. Recalculate cluster distances according to the selected linkage method.
6. Repeat until one hierarchy remains.
7. Use the dendrogram to select a useful number of clusters.

5. Complete practical code

```
# =====
# AGGLOMERATIVE HIERARCHICAL CLUSTERING
# =====

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage

# 1. Read dataset
df = pd.read_csv("customers.csv")
print("Original Dataset:")
print(df)

# 2. Select features
X = df[["Annual_Income", "Spending_Score"]]

# 3. Standardize data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
# 4. Create linkage matrix
linked = linkage(X_scaled, method="ward")

# 5. Plot dendrogram
plt.figure(figsize=(10, 6))
dendrogram(linked, labels=df["Customer"].values)
plt.title("Hierarchical Clustering Dendrogram")
plt.xlabel("Customers")
plt.ylabel("Distance")
plt.show()

# 6. Apply Agglomerative Clustering
hc = AgglomerativeClustering(
    n_clusters=2,
    linkage="ward"
)
clusters = hc.fit_predict(X_scaled)

# 7. Add cluster labels
df["Cluster"] = clusters

# 8. Display result
print("\nFinal Dataset:")
print(df)

# 9. Visualize clusters
plt.figure(figsize=(8, 5))
plt.scatter(
    X["Annual_Income"],
    X["Spending_Score"],
    c=clusters,
    s=100
)
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("Hierarchical Clustering")
plt.show()
```

6. Linkage methods

Linkage	Meaning
Single	Minimum distance between two clusters
Complete	Maximum distance between two clusters
Average	Average pairwise distance between clusters
Ward	Merges clusters while minimizing increase in within-cluster variance

7. Cutting the dendrogram

The vertical axis shows the linkage distance. Draw an imaginary horizontal line across the dendrogram. The number of major branches crossed by that line gives the resulting number of clusters. A useful rule of thumb is to look for a large vertical gap between merge heights and place the cut inside that gap.

```
from scipy.cluster.hierarchy import fcluster

# Cut the dendrogram at a chosen distance
clusters = fcluster(
    linked,
    t=5,
    criterion="distance"
)

df["Cluster"] = clusters
print(df)
```

8. Distance metrics

Euclidean distance: straight-line distance.

Manhattan distance: sum of absolute differences across dimensions. In hierarchical clustering, the exact distance calculation and linkage method should be considered separately.

9. Important notes

- Scaling is recommended for features measured on different scales.
- The cluster numbers themselves have no inherent meaning.
- Dendrogram height represents the distance/dissimilarity at which merges occur.
- Agglomerative clustering is bottom-up: start with individual observations and merge.

10. Student exercise

Try Single, Complete, Average, and Ward linkage on the same dataset. Plot each dendrogram, compare the merge structure, choose a reasonable cut height, and explain why the resulting clusters make sense.